

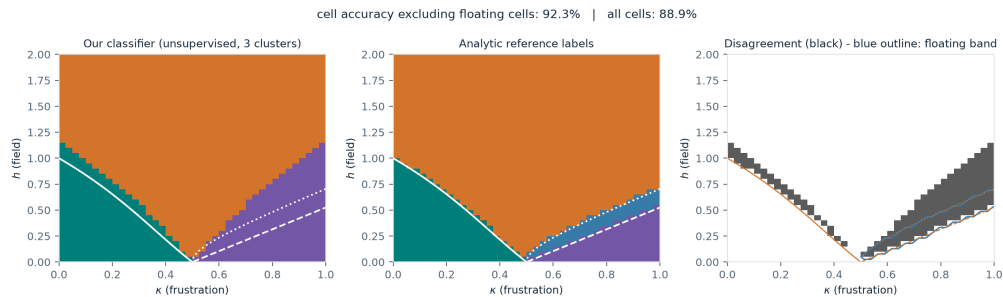
PHASE HUNTER

Team Popeyes - solo: Tanish Singh Rajpal | Q-SITE 2026 Scientific Track

A game about finding a quantum phase boundary with a finite measurement budget. Players scan a hidden map, interpret two correlators and draw a border. The contribution is a reproducible investigation of finite-size frustration, gate noise and measurement strategy, made playable in a browser.

1. Model, observables and classification

The periodic ANNNI ring has $H = -\sum ZZ(1) + \kappa \sum ZZ(2) - h \sum X$. We scan κ in $[0,1]$ and h in $[0,2]$. C1 and C2 average Z correlations over all sites. Three seeded k-means clusters classify ferromagnetic, paramagnetic and antiphase regions. The analytic floating band is excluded from the primary metric; it is not independently detected.



The $N=8$, 40×40 exact PennyLane grid agrees with the qualitative analytic reference on 92.28% of non-floating cells (88.94% of all cells). NumPy and PennyLane energies/correlators agree to about $3e-14$. Finite-size label agreement is not an exact thermodynamic transition location.

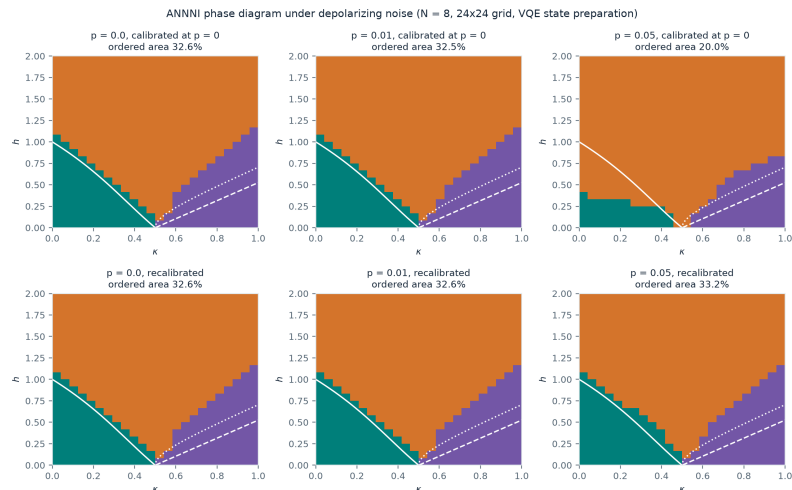
2. A ring-size trap: period-four order

A perfect $++--$ stripe closes only when N is divisible by four. On $N=6$, C2 cannot be below $-1/3$ for any Z-basis bitstring, so averaging quantum outcomes cannot recover the -1 stripe signature. The same classifier therefore behaves very differently even without gate noise.

Ring N	C2 at $\kappa=0.8, h=0.1$	24 x 24 label agreement
6	-0.332	43.9%
8	-0.996	91.9%
12	-0.996	95.1%

This is finite-ring commensuration and a classifier limitation, not disappearance of the infinite-model antiphase. The 24×24 size comparison uses the same grid, reference mask and three-cluster rule; its $N=8$ score differs from the denser 40×40 headline result.

3. Noise changes the signal and the inferred map



Gate noise p	Ordered area: fixed clean rule	Ordered area: refitted rule
0.0	32.64%	32.64%
0.01	32.47%	32.64%
0.05	19.97%	33.16%

The 24×24 $N=8$ scan uses four-layer VQE fits on default.qubit, then identical parameters on default.mixed with target depolarization after each CNOT. Mean preparation-energy error is 0.149, maximum 0.323; 443/576 points exceed 0.05. Comparisons across p share the prepared state.

At $p=0.05$ the fixed classifier calls about 39% less area ordered. Recalibration recovers most of that area, with small boundary changes. This demonstrates classifier sensitivity, not invariant physical transitions. Antiphase and ferromagnetic interiors lose about 47% and 40% of their signals; the $N=12$ cuts reproduce the ordering (0.512 vs 0.615). Likely mechanism: the antiphase lives on the distance-2 correlator, with more gates between the compared spins, so the same per-gate error costs it more. Both sizes agree with this handout prediction, but isolating distance would need a third distance or a depth sweep.

Paired finite-shot test (200 seeds, same full-register observations): at $p=0.05$ and 20 shots/point, recalibration raises agreement with the noiseless prepared-map labels from 85.4% to 93.7%. At $p=0.01$ it falls from 98.5% to 97.6%, showing that refitting is useful conditionally rather than automatically. This is transductive map inference, not state repair or held-out generalization.

4. Larger-system check: $N=12$ with noise

Cut (κ)	$p=0.05$ attenuation	Fixed / rescaled shift in h
0.2	0.615	-0.131 / +0.037
0.8	0.512	-0.383 / +0.036

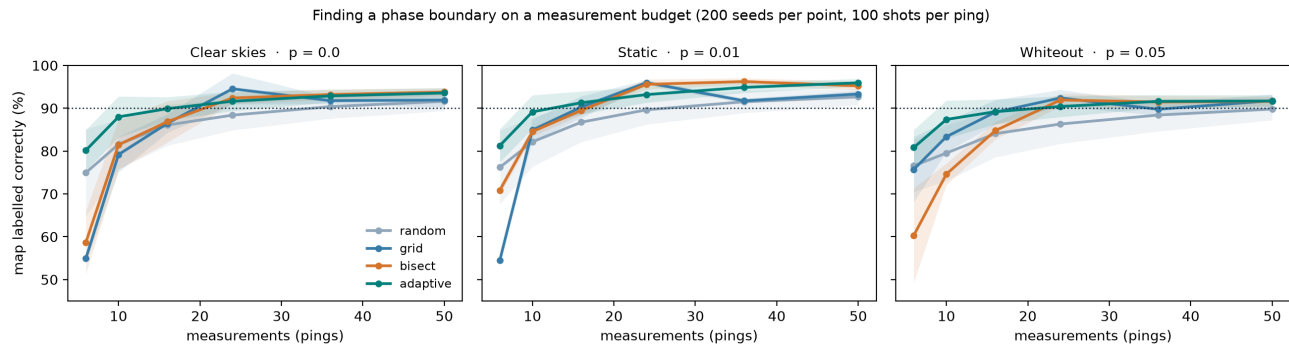
Two nine-point cuts use $h=0.1$ to 1.3 (step 0.15), four-layer VQE and the same noise placement. Rescaling brings crossings within 0.037 of their $p=0$ positions; sparse threshold interpolation is not a phase-transition proof. Preparation-energy errors are 0.186 mean and 0.474 maximum. Full data and plots are in the notebook.

5. Independent diagnostic and mitigation

Fidelity susceptibility in one fixed translation-invariant, spin-flip-even sector ($h=0$ excluded) locates peaks a mean 0.046 in h from the analytic reference, at spacing 0.051. Its mean gap to clustering is 0.128; the methods do not agree within one grid step on average.

Two-point exponential zero-noise extrapolation reduces $C1/C2$ errors from 0.120/0.179 to 0.0019/0.0020, relative to the same noiseless variational state. These infinite-shot results do not assess finite-shot error amplification or exact-ground-state recovery.

6. Shared readout and a fair measurement budget



One shot measures the whole register in Z. The same bitstring supplies both C1 and C2, each averaged over N sites. A ping draws 100 joint outcomes from the exact computational-basis distribution of its prepared state. Multinomial sampling preserves physical bounds, per-shot variance and covariance; it neither splits shots nor reuses clean-state widths for noisy states. Recovered N=8 fits are checked against every archived mean and energy.

Random, grid, bisection and adaptive methods use exactly 6, 10, 16, 24, 36 or 50 pings, with 200 seeds per budget. A fixed $\max(C1, -C2) > 0.46$ rule supplies votes; three-nearest weighted voting reconstructs the map. Scores use stage cluster labels, excluding reference floating cells. Shading is one standard deviation across seeds, not a confidence interval.

16 pings	Random	Grid	Bisect	Adaptive
Clear skies	86.1%	86.5%	86.9%	89.9%
Static	86.7%	90.3%	89.5%	91.3%
Whiteout	84.1%	89.1%	84.8%	89.2%

Clear skies: first tested 90% crossing - random: 36, grid: 24, bisect: 24, adaptive: 24.

Static: first tested 90% crossing - random: 36, grid: 16, bisect: 24, adaptive: 16.

Whiteout: first tested 90% crossing - random: not reached, grid: 24, bisect: 24, adaptive: 24.

5 tested means lie within two standard errors of 90%; their crossing budgets are provisional. Cross-stage comparisons also mix state preparation and classification, so they do not isolate a universal noise overhead.

7. Play, reproduce and extend

Open [game/index.html](#). Practice/Hunter provide 90/60 energy. Weak/solid/deep scans use 20/100/500 whole-register shots for 1/2/4 energy. The AI visibly scans the same map and budget; its five-handle interpolation differs from the study reconstruction. Twelve famous GIFs, personal recordings and short lessons reward meaningful events. Retry keeps the map and AI target.

Limitations: finite rings, heuristic classifiers, unresolved floating phase, imperfect VQE fits and no hardware or quantum-advantage claim. Team Popeyes is solo: Tanish Singh Rajpal. The presentation video remains to be recorded. README.md contains the current UI captures, reproducible commands and links to validation data.

Sources: Quantum Coalition QSITE-2026 Scientific Track handout/starter kit; PennyLane ANNNI tutorial, computational-basis probabilities and DepolarizingChannel documentation. See README.md for direct links.